Microsoft Learn — Azure documentation: https://learn.microsoft.com/azure/

Microsoft Learn — AZ-900 study guide: https://learn.microsoft.com/en-us/credentials/certifications/resources/study-guides/az-900

Official exam reference

Cloud behaviour depends on service type, configuration, region, and workload. Avoid absolute statements such as “always,” “never,” or “automatically” unless the documentation explicitly supports them. The notes below are learning material, not a substitute for current official documentation.

Accuracy note:

4. Verify service details, limits, and exam objectives against the current Microsoft Learn study guide before the exam.

3. Treat “VERY IMPORTANT” and “MOST ASKED” labels as revision cues, not guarantees about what will appear on the exam.

2. Practise distinguishing similar services (for example, VPN Gateway vs ExpressRoute, Azure Monitor vs Service Health, and RBAC vs Azure Policy).

1. Read one topic, then explain it in your own words without looking.

How to use these notes

Beginner-friendly revision notes for Microsoft Azure Fundamentals (AZ-900). These notes explain concepts in plain language, use examples and exam-style reminders, and are intended to be studied alongside Microsoft Learn.

Purpose

Study guide and revision notes

Azure-900 Fundamentals

Main pillars:

☁️ Compute

🌐 Networking

🗄️ Databases

💾 Storage

🤖 AI & ML

📡 IoT

What is Azure?
Microsoft Azure is a cloud computing platform that provides services like:

- Virtual Machines
- Networking
- Storage
- Databases
- Security
- AI services

It helps companies build and manage applications without maintaining physical hardware.

Why is Azure Important?

- Reduces hardware cost
- Scalable infrastructure
- Global availability
- Faster deployment
- Secure cloud environment
- Used by enterprises worldwide

Where is Azure Used?

- Hosting websites
- Cloud servers
- Enterprise applications
- Data storage
- Backup solutions
- Networking infrastructure
- Identity management

☁️ **Compute -** Services that provide processing power. **Ex –** VM, Containers, App services
Why important?
Applications need servers to run.

🌐 **Networking** - Services that connect cloud resources securely. **Ex –** VNet, Load balancer, DNS, VPN
Why important?
Resources need communication.

🗄️ **Databases** - Managed database services in Azure. **Ex –** Azure SQL, Cosmos SQL, PostgreSQL
Why important?
Applications need structured data storage.

💾 **Storage -** Services used to store files/data.
Ex – Blob Storage, File Storage
Why important?
Applications generate/store huge data.

Cloud Computing - delivery of IT services over the internet instead of using physical hardware locally.

These services include:

- Compute (servers)
- Storage
- Networking
- Databases
- Security
- AI/ML services

Traditional Infrastructure	Cloud Computing
❌ Requires buying physical servers	✅ On-demand virtual servers
❌ Must set up and manage own data centers	✅ Managed by cloud providers
❌ High (hardware, cooling, staff)	✅ Reduced (provider handles infrastructure)
❌ Long (procurement, installation, configuration)	✅ Faster (ready-to-use resources)
❌ Limited, needs new hardware	✅ Elastic scaling up/down instantly
❌ High upfront investment	✅ Pay-as-you-go, reduced hardware cost
❌ Limited to local setup	✅ Global availability across regions

🌎 **Where is Cloud Computing Used?**

- Web hosting
- Enterprise applications
- Online storage
- Backup & disaster recovery
- Streaming services
- Banking systems
- E-commerce
- AI/ML workloads

Key Benefits of Cloud Computing:

1️⃣ **Scalability:** Increase/decrease resources based on demand.
2️⃣ **High Availability:** Services remain available with minimal downtime.
3️⃣ **Global Reach:** Deploy applications in multiple regions worldwide.
4️⃣ **Pay-as-you-go:** Only pay for resources you use.
5️⃣ **Faster Deployment :** Resources can be created within minutes.

Shared Responsibility Model

In cloud computing, responsibilities are shared between:

- Cloud Provider (Microsoft Azure)
- Customer/User

❓ **Why is it Important?** Defines:
✓ Who manages what
✓ Security responsibilities
✓ Maintenance responsibilities
✓ Infrastructure ownership

🏢 On-Premises (Traditional Datacentres) - Customer manages EVERYTHING.
Including:

- Physical servers
- Networking
- Operating systems
- Applications
- Security
- Databases

☁️ In Cloud -
Azure manages:
✓ Physical datacenter
✓ Hardware
✓ Physical network
✓ Power/cooling
Customer still manages:
✓ Data
✓ Accounts/users
✓ Access permissions

Customer is always responsible for:

- Data/information
- User accounts
- Identity & access
- Devices connecting to cloud

🏆 **Shared Responsibility by Service Model:**
☁️ **IaaS** (Infrastructure as a Service)
⚙️ **PaaS** (Platform as a Service)
💻 **SaaS** (Software as a Service)

☁️ **IaaS:** More customer control/responsibility
Ex- Azure VM
Azure Manages:

- Physical infrastructure
- Datacenter
- Hardware

Customer manages:

- OS
- Applications
- Network controls
- Data

⚙️ **PaaS:** Focus on application
Ex – Azure App Services
Azure manages:

- Infrastructure
- Operating system
- Runtime/platform

Customer manages:

- Applications
- Data
- Access

💻 **SaaS:** Least customer responsibility
Ex – Microsoft 365
Azure/provider manages almost everything
Customer mainly manages:

- Users
- Access
- Data

Responsibility decreases:

On-Prem → IaaS → PaaS → SaaS

Cloud models

Define how cloud infrastructure is deployed and managed.
Main models:

- **Public** Cloud
- **Private** Cloud
- **Hybrid** Cloud

☁️ **Public Cloud:** Cloud services provided over the internet by third-party providers like Microsoft Azure. **Ex -**

- Microsoft Azure
- AWS
- Google Cloud

Why important?
No need to buy/manage physical infrastructure.

Key Features:
✓ Pay-as-you-go
✓ Fast deployment
✓ Scalable
✓ Managed by provider

Advantages:

- Low cost
- Quick provisioning
- No hardware maintenance

Disadvantages:

- Less control
- Shared environment

👉 **Public Cloud = Shared cloud managed by provider**

🏢 **Private Cloud:** Cloud infrastructure dedicated to a single organization.

Why important?
Provides more control and security.

Key Features:
✓ Dedicated resources
✓ More security/control
✓ Single organization use

Advantages:

- Better control
- Better customization
- Strong security/compliance

Disadvantages:

- Expensive
- Hardware maintenance required

👉 **Private Cloud = Dedicated cloud for one organization**

🔁 **Hybrid Cloud:** Combination of Public + Private cloud connected together.

Why important?

Allows flexibility between environments.

Key Features:
✓ Mix of both clouds
✓ Flexible deployment
✓ Better control over sensitive data

Advantages:
- Flexibility
- Cost optimization
- Better compliance

Common Use Case:
Sensitive data stays in private cloud while applications run in public cloud.

👉 **Hybrid Cloud = Public + Private together**

🌐 **Multicloud:** Using multiple cloud providers together.

Example:
- Azure + AWS
- Azure + Google Cloud

Why used?
- Different provider strengths
- Migration scenarios
- Redundancy/flexibility

👉 **Multicloud = Multiple cloud providers**

⭐ **Azure Arc:**

What is it?
Azure service used to manage:
- Hybrid environments
- Multicloud environments from one place.

👉 **Azure Arc = Unified cloud management**

⭐ **Azure VMware Solution:**
Allows VMware workloads to run inside Azure.

Why important?

Helps companies migrate existing VMware infrastructure to Azure cloud.

Cloud Model	Main Feature
Public Cloud	Low cost & scalable
Private Cloud	High control/security
Hybrid Cloud	Maximum flexibility
Multicloud	Multiple providers

You pay only for the resources you consume.

⭐ **Core Idea:**
👉 Use resources when needed
👉 Stop paying when not needed

🏢 **CapEx (Capital Expenditure):** Upfront spending on physical infrastructure.

Examples:
- Buying servers
- Datacenter setup
- Networking hardware

👉 **CapEx = Buying infrastructure**

☁️ **OpEx (Operational Expenditure):** Paying for services over time based on usage.

Examples:
- Monthly cloud billing
- VM usage charges
- Storage usage charges

👉 **OpEx = Paying as you use**

☁️ **Benefits of Consumption-Based Model:**

1️⃣ **No upfront hardware cost**
No need to purchase physical servers.

2️⃣ **Scalability**
Resources can increase when demand increases.

3️⃣ Elasticity
Resources can decrease when demand decreases.

4️⃣ Cost Optimization
Pay only for actual usage.

📈 Scalability	🔄 Elasticity
What is it? Ability to increase resources to handle higher demand.	**What is it?** Ability to automatically increase AND decrease resources based on demand.
Example Adding more VMs during high traffic.	**Example** Traffic decreases → extra VMs removed automatically.
Recall Point 👉 Scalability = Grow resources	**Recall Point** 👉 Elasticity = Dynamic scaling up/down

Cloud computing is mainly:
👉 OpEx model

What does cloud reduce?
👉 CapEx costs

What does elasticity mean?
👉 Resources automatically scale up/down based on demand

What does scalability mean?
👉 Ability to increase resources when needed

Benefits of using cloud services:

High Availability & Scalability:

High Availability (HA): Ability of a system/service to remain available with minimal downtime.

❓ Why is it Important?
Businesses need applications/services available continuously.
Downtime can cause:
- Financial loss
- User dissatisfaction
- Service interruption

👉 High Availability = Maximum uptime

📊 SLA (Service Level Agreement):
Commitment from cloud provider about expected uptime.

SLA	Approx Downtime/Year
99%	~3.65 days
99.9%	~8.76 hours
99.99%	~52 minutes

❓ Why is it Important?
Applications may experience:
- High traffic
- Increased workload
- Seasonal demand spikes

Cloud can quickly adjust resources.

👉 Scalability = Adjust resources according to demand

📈 Types of Scaling

1️⃣ Vertical Scaling (Scale Up/Down)
Increase/decrease power of existing resource.

Example:
- More CPU
- More RAM

to same VM/server

👉 Vertical Scaling = Bigger machine

2️⃣ Horizontal Scaling (Scale Out/In)
Add/remove multiple resource instances.

Example:
Add more VMs/containers.

👉 Horizontal Scaling = More machines

🧠 Vertical Scaling	🧠 Horizontal Scaling
Increase power	Increase number
Bigger VM	More VMs
Scale Up	Scale Out

☁️ **Benefits of Scalability:**
✓ Better performance
✓ Handle traffic spikes
✓ Cost optimization
✓ Flexible resource usage

💔 **Real-World Example:**
E-commerce website during festival sale
What is High Availability?
👉 Ensuring maximum uptime with minimal downtime

What is SLA?
👉 Uptime guarantee from provider

Which scaling adds more CPU/RAM?
👉 Vertical Scaling

Which scaling adds more VM instances?
👉 Horizontal Scaling

Which scaling is called “Scale Out”?
👉 Horizontal Scaling

Benefits of reliability and predictability in the cloud:

Reliability: Ability of a cloud system to recover from failures and continue functioning.

❓ **Why is it Important?**
Systems can fail because of:
- Hardware issues
- Network problems
- Regional outages
- Traffic spikes

Cloud systems are designed to remain operational.

👉 **Reliability = System continues working even during failures**

☁️ **How Cloud Improves Reliability:**
Cloud uses:
✓ Multiple regions
✓ Distributed infrastructure
✓ Failover mechanisms
✓ Redundancy

🌎 **Example:**
If one Azure region fails:
Another region can continue serving users.

👉 **Reliability = Recover from failure & continue functioning**

🔁 **Failover:** Automatic switching to another working resource/region during failure.
👉 **Failover = Backup system takes over automatically.**

📈 **Predictability:** Ability to predict:
- Performance
- Resource usage
- Costs

before deployment.

⭐ **Two Types of Predictability**

1️⃣ **Performance Predictability:**
Ensures applications perform consistently.

Supported By:
✓ Autoscaling
✓ Load balancing
✓ High availability

👉 **Performance Predictability = Stable application performance**

2️⃣ **Cost Predictability:** Ability to estimate and monitor cloud spending.

Supported By:
✓ Pricing calculator
✓ Monitoring tools
✓ Usage analytics

👉 **Cost Predictability = Estimate cloud cost in advance**
⚖️ **Load Balancing:** Distributes traffic/workload across multiple resources.

Why important?
Prevents one server from becoming overloaded.

👉 **Load Balancer = Distributes traffic**

🔁 **Autoscaling:** Automatically adds/removes resources based on demand.

👉 **Autoscaling = Automatic resource adjustment**

Reliability means:
👉 Recovering from failures while continuing operation

Predictability focuses on:
👉 Performance + Cost

What supports performance predictability?
👉 Autoscaling + Load Balancing + High Availability

What supports cost predictability?
👉 Pricing calculators + Monitoring tools

❤️Example:
Streaming platform
Reliability: If one server/region fails → service still works.
Predictability:
Company estimates:

- Traffic demand
- Server requirements
- Monthly cloud cost

before deployment.

Security & Governance in Cloud:

Security:
DDoS protection: Protection against Distributed Denial of Service attacks.

👉 **DDoS = Flood attack on services**

⚙️ **Patching & Maintenance:** Cloud providers automatically update infrastructure/software.

Why important?
Keeps systems secure and stable.

👉 Cloud reduces manual maintenance

📔 **Governance in Cloud:** Set of rules, policies, and standards for managing cloud resources properly.

❓ **Why is it Important?**
Ensures:
✓ Compliance
✓ Standardization
✓ Cost control
✓ Security consistency

👉 **Governance = Managing cloud resources according to rules/policies**

📑 **Governance Benefits**
✓ Resource standardization
✓ Policy enforcement
✓ Compliance monitoring
✓ Audit support
✓ Centralized management

👉 **Governance = Rules & management for cloud resources**

🔍 **Compliance:** Following legal, regulatory, and organizational standards.

Example:

- Data protection laws
- Security standards
- Industry regulations

👉 **Compliance = Following required standards/rules**

🔁 **Auditing:** Monitoring and checking cloud resources for policy/security violations.

👉 **Auditing = Tracking & reviewing resource activities**

Security in cloud includes:
👉 DDoS protection + patching + access control

Governance focuses on:
👉 Policies, standards, compliance, management

Compliance means:
👉 Following legal/regulatory requirements

Why is governance important?

👉 Keeps cloud environment secure, standardized, and controlled

Benefits of manageability in the cloud:

Manageability: Ability to efficiently manage, monitor, automate, and maintain cloud resources.

❓ **Why is it Important?**
Cloud environments can contain:

- Many servers
- Databases
- Networks
- Applications

Manageability helps control them efficiently.

👉 **Cloud simplifies resource management through automation & monitoring**

☁️ **Two Types of Manageability:**

1️⃣ **Management OF the Cloud:** How cloud platform automatically manages resources.

Features:
✓ Autoscaling
✓ Health monitoring
✓ Templates
✓ Alerts

👉 **Cloud automatically manages infrastructure/resources**
🔄 **Autoscale:** Automatically increases/decreases resources based on demand.

👉 **Autoscale = Automatic scaling**

💓 **Health Monitoring**
Cloud continuously checks resource health/status.

Why important?
Detects failures/issues quickly.

👉 **Health Monitor = Resource status checking**

📑 **Templates:** Predefined configurations for resource deployment.

Why important?
Enables consistent/repeatable deployments.

👉 **Templates = Reusable deployments**

🚨 **Alerts:**
Notifications triggered based on metrics/events.

Example:
High CPU usage alert.

👉 **Alerts = Real-time notifications**

2️⃣ **Management in the Cloud:** Ways users/admins manage cloud resources.

⭐ **Management Tools**

🖥️ **Azure Portal:** Web-based graphical interface for managing Azure resources.

👉 **Portal = GUI management**

💻 **CLI (Command Line Interface):** Text-based command tool for Azure management.

👉 **CLI = Command-based management**

🔌 **APIs:** Programmatic access to Azure services.

👉 **APIs = Automation/programmatic control**

⚡ **PowerShell:** Scripting tool used for automation and management.

👉 **PowerShell = Automation scripting**

What does manageability mean?

👉 Efficient management & automation of cloud resources

What is Azure Portal?
👉 Web-based GUI management tool

What is CLI used for?
👉 Command-line management

What are templates used for?
👉 Repeatable deployments

What are alerts used for?
👉 Real-time monitoring notifications

Sustainability considerations in the cloud:

Sustainability: Using cloud resources efficiently to reduce waste, energy usage, and unnecessary resource consumption.

❓ **Why is it Important?**
Traditional datacenters often:
❌ Waste power/resources
❌ Run unused servers
❌ Overprovision infrastructure
Cloud improves efficiency through dynamic resource management.

👉 **Use only the resources you actually need**

☁️ **Sustainability Practices:**

1️⃣ **Right-Size Resources:** Choose appropriate resource size based on actual demand.

Example:
Avoid oversized VMs.
👉 **Right-size = Match resources to workload**

2️⃣ **Automate Resource Management**
Automatically scale/shutdown unused resources.

Example:
Auto shutdown development VM at night.

👉 **Automation reduces waste**

3️⃣ **Optimize Resources**: Use efficient services/configurations.

👉 **Optimization = Efficient cloud usage**

4️⃣ **Monitor Usage:** Track resource usage and spending continuously.

👉 **Monitoring helps reduce unnecessary consumption**

🌱 **Benefits of Sustainability**
✓ Reduced energy usage
✓ Better resource utilization
✓ Lower costs
✓ Reduced waste
✓ Efficient scaling

Sustainability focuses on:
👉 Efficient resource usage

What helps sustainability?
👉 Right-sizing + Automation + Monitoring + Optimization

Why is cloud more sustainable?
👉 Resources scale dynamically instead of remaining idle

❤️‍🩹 **Example:**
Company development servers
Instead of running 24/7:
✓ Auto shutdown at night
✓ Scale resources based on demand
✓ Monitor usage trends
This reduces:

- Cost
- Power consumption
- Resource waste

Cloud service types:

Infrastructure as a Service (IaaS): IaaS provides cloud-based infrastructure resources such as:

- Virtual Machines
- Storage
- Networking

through the internet.

❓ **Why is it Important?**
Companies can:
✓ Use cloud infrastructure
✓ Avoid buying physical servers
✓ Scale quickly
✓ Maintain greater control over systems

👉 **You rent infrastructure from cloud provider**

☁️ **In IaaS**
Azure/Cloud Provider manages:
✓ Physical servers
✓ Datacenter
✓ Hardware
✓ Physical networking

Customer/User manages:
✓ Operating System
✓ Applications
✓ Data
✓ Runtime
✓ Security configurations
✓ Network settings

👉 **IaaS = Maximum customer control in cloud + maximum flexibility and the largest operational responsibility**

🔧 **What You Manage in IaaS**
✓ OS installation/configuration
✓ Software installation
✓ Networking configuration
✓ Database configuration
✓ Security settings

🌟 **Common IaaS Services**
Examples

- Azure Virtual Machines
- Azure Virtual Network
- Azure Storage

⭐ **Common Use Cases**

1️⃣ **Lift-and-Shift Migration:** Moving existing on-premises applications directly to cloud infrastructure.

👉 **Lift-and-Shift = Move existing infrastructure to cloud**

2️⃣ **Testing & Development:** Quickly create/remove environments for testing.

🌟 **Benefits of IaaS:**
✓ High flexibility
✓ High control
✓ Scalable infrastructure
✓ Faster deployment
✓ No physical hardware management

⚠️ **Important Limitation:**
You still manage:

- OS updates
- Security patches
- Applications
- Configurations

So responsibility is HIGH.

Which cloud model provides MOST control?
👉 IaaS

In IaaS, who manages Operating System?
👉 Customer/User

In IaaS, who manages physical infrastructure?
👉 Cloud Provider

Example of IaaS in Azure?
👉 Azure Virtual Machine
👉 IaaS = Rent hardware, manage everything else

💔 Example:
Company creates Azure VM
Azure handles:
✓ Datacenter
✓ Hardware
✓ Power/networking
Company handles:
✓ Linux installation
✓ Applications

✓ Firewall rules
✓ User accounts

Platform as a Service (PaaS): PaaS provides a cloud platform/environment for developing, running, and managing applications.

❓ **Why is it Important?**
Developers can focus on:
✓ Application development
✓ Business logic
✓ Data
without managing infrastructure.

👉 **You build applications without managing servers/OS**

☁️ **In PaaS:**
Azure/Cloud Provider manages:
✓ Physical infrastructure
✓ Servers
✓ Storage
✓ Networking
✓ Operating System
✓ Middleware
✓ Runtime

Customer/User manages:
✓ Applications
✓ Data
✓ Some access/security settings

👉 **PaaS = Focus on application, not infrastructure**

⚙️ **What is Middleware?**
Software layer between OS and applications.
Examples:
- Web servers
- Application frameworks
- Runtime environments

🌟 **Common PaaS Use Cases:**

1️⃣ **Development Framework:**
Developers quickly build cloud applications using built-in tools/services.

Benefits:
✓ Faster development
✓ Scalability
✓ Reduced infrastructure management

2️⃣ **Analytics & Business Intelligence:**
Cloud platform provides tools for:
- Data analysis
- Reporting
- Insights

🌟 **Benefits of PaaS:**
✓ Reduced management overhead
✓ Faster application deployment
✓ Built-in scalability
✓ Automatic updates/patching
✓ Simplified development environment

⚠️ **Important Limitation:**
Less infrastructure control compared to IaaS.

In PaaS, who manages Operating System?
👉 Cloud Provider

In PaaS, who manages Applications & Data?
👉 Customer/User

Main purpose of PaaS?
👉 Application development without infrastructure management

Which provides more control?
👉 IaaS

Which provides less management overhead?
👉 PaaS

❤️‍🩹**Example:**
Developer creates web application
With PaaS:
Azure handles:
✓ Servers
✓ OS updates
✓ Runtime
✓ Scaling
Developer focuses on:
✓ Application code
✓ Features
✓ Data

Software as a Service (SaaS): least flexible

SaaS provides fully developed software/applications over the internet.

Users simply use the application without managing infrastructure or platform.

❓ Why is it Important?

Users can:

✓ Access applications quickly
✓ Avoid infrastructure management
✓ Avoid software maintenance
✓ Use applications from anywhere

👉 You use the software — provider manages everything else

☁️ In SaaS:

Azure/Cloud Provider manages:

✓ Infrastructure
✓ Servers
✓ Networking
✓ Storage
✓ Operating System
✓ Runtime
✓ Applications
✓ Updates & maintenance

Customer/User manages:

✓ Data
✓ Users
✓ Access permissions

👉 SaaS = Fully managed software

🌟 Common SaaS Examples:

- Outlook
- Microsoft 365
- Teams
- Word
- Excel Online

Other Examples

- Gmail
- Google Docs
- Dropbox
- Salesforce

🌟 Common SaaS Use Cases:

✓ Email & messaging
✓ Productivity applications
✓ Collaboration tools
✓ Finance/accounting software

🌟 Benefits of SaaS:

✓ Easy to use
✓ Minimal setup
✓ No infrastructure management
✓ Automatic updates
✓ Accessible from anywhere

⚠️ Important Limitation:

Least infrastructure control for customer.

Which cloud model gives LEAST customer responsibility?
👉 SaaS

In SaaS, who manages applications?
👉 Cloud Provider

In SaaS, what does customer mainly manage?
👉 Data + Users + Access

Example of SaaS?
👉 Microsoft 365